

Decision Record and Architecture Hypotheses

Every arbitration taken during the founding design sessions of 22–23 September 2026, the schemas of the kernel and the userland they produce, and the hypotheses the roadmap must validate.

Project VerticalAI — self-improving, peer-to-peer, local-first orchestration runtime for businesses
Founder Eric LTX · **Repository** github.com/EricLtx/VerticalAI · **State** SP0 delivered (tag `sp0`), CI green on Windows / Linux / macOS
Sources design spec `docs/superpowers/specs/2026-09-22-verticalai-design.md`; SP0 plan; literature brief (32 sources read in full); adversarial design review (47 findings, 40 confirmed)

Status legend used throughout: decided arbitration taken by the founder · literature supported by published evidence · to validate hypothesis with a named test · known risk accepted with mitigation

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1. Vision and value ladder

VerticalAI is a **self-improving, peer-to-peer, local-first orchestration runtime** for businesses — from a one-person company to the employees of a large enterprise. The business owns everything it accumulates; that *userland* is movable, replicated across the business's own machines with no cloud dependency, and shareable within a trade through a curated federation. Cloud models are optional *arches*, never a dependency.

Analogy kept internally: "Linux for AI". A small kernel inspired by micro-architecture — orchestration (pipeline/scheduler), registers, locks — with everything else in userland. Models are the *instruction set*; the kernel's task-state IR is the common intermediate representation that lets heterogeneous models cooperate. **Externally** the product is described as an orchestration runtime, not an "AI operating system" (CRA Annex III wording; see §14).

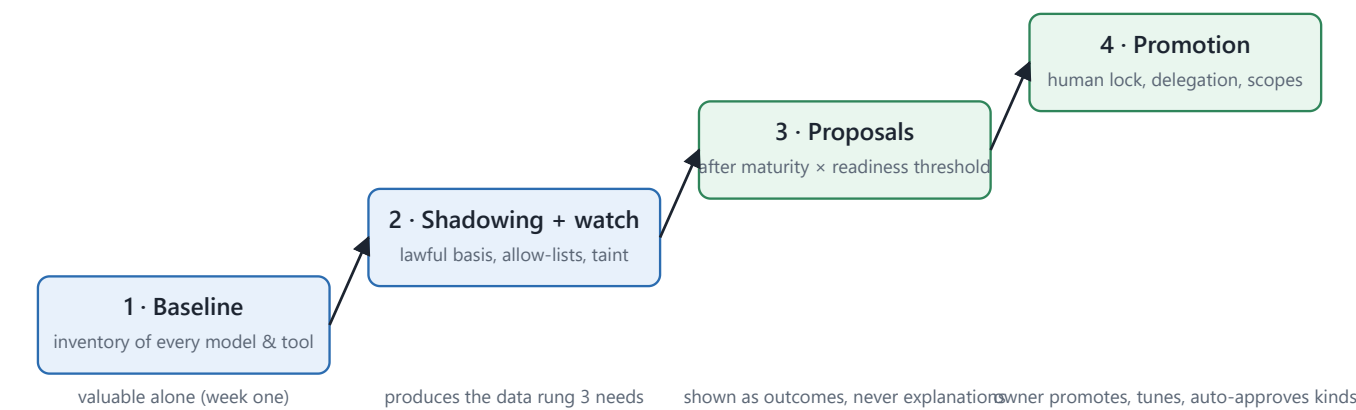


Figure 1 — The value ladder. Each rung produces the data the next needs; the first rung is useful before anything has been learned.

Arbitration	Content	Status
First trade	Freelancers and SMEs selling services or products, in any craft (design, consulting, content, AI-native platforms). First user: the founder's one-person company (sole-trader mode).	decided
Commercial model	Commercial first, to build activity; the kernel is open-sourced later to democratise the commercial offering (userland, federation, support).	decided
Non-functional requirements	Cybersecurity and privacy by design; user-centric; organic and autonomous construction <i>within gates</i> ; pedagogic (show outcomes); compliance by design (AI Act, GDPR, e-Privacy, NIS 2, CRA, PLD); easy and low-cost deployment; movable and traceable configuration; simple migration; ongoing technology watch.	decided

2. System topology

One shared userland per business, replicated across its own nodes; a phone acts as a thin node for human approvals; a headless node may host arches and relay traffic; cloud arches are reached only through the kernel and only within clearance. Businesses in the same trade share modules through a curated federation.

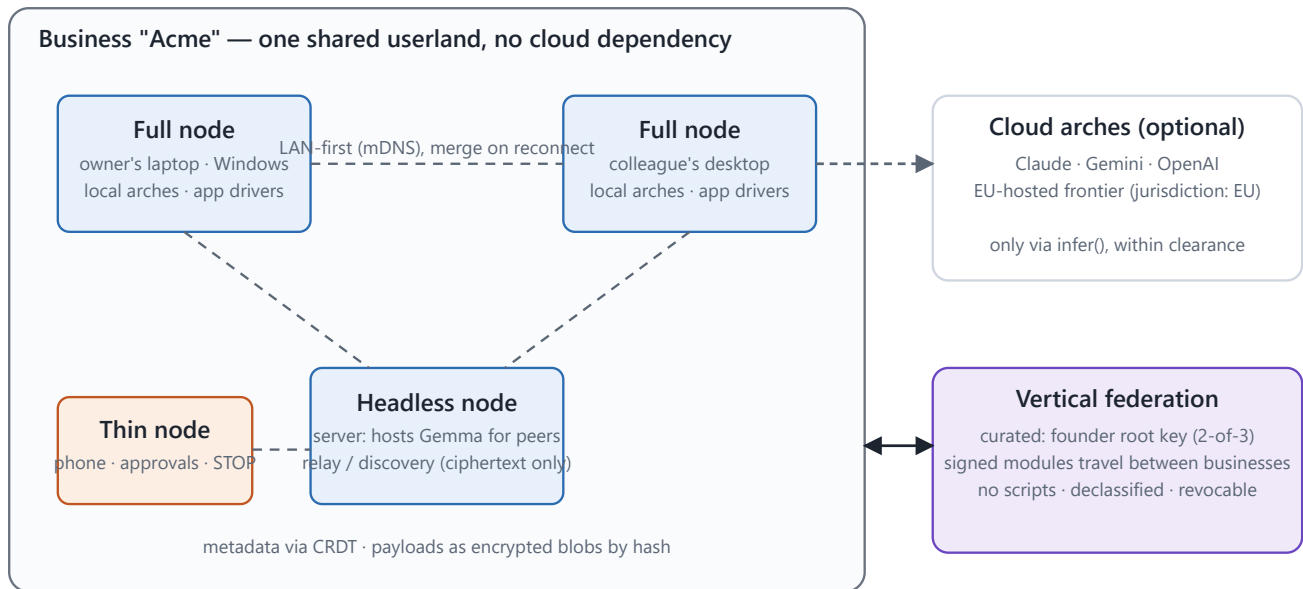


Figure 2 — Nodes, the business mesh, optional cloud arches and the federation. Node classes differ only by manifest.

Node class	Hosts arches	Drives applications	Role
Full	yes	yes	The owner's laptop or desktop; runs the kernel daemon natively (Windows first, Linux, macOS)
Headless	yes	no	Server; also relay / discovery role for the business or the trade (sees ciphertext only)
Thin	no	no	Phone: identity, channel, STOP, human approvals; persists no registers and no secrets

3. Kernel / userland boundary

The kernel **owns** only mechanism: scheduling, the registers (task-state IR), the lock surface, STOP, principals, the arch layer, storage tiering, the ledger, replication mechanics, and the interceptor chain. It **refuses** to own any domain knowledge, skill, business memory, policy, specific model, application driver, channel or user interface. Anything policy-shaped is a userland object the kernel reads and enforces but never writes on its own initiative.

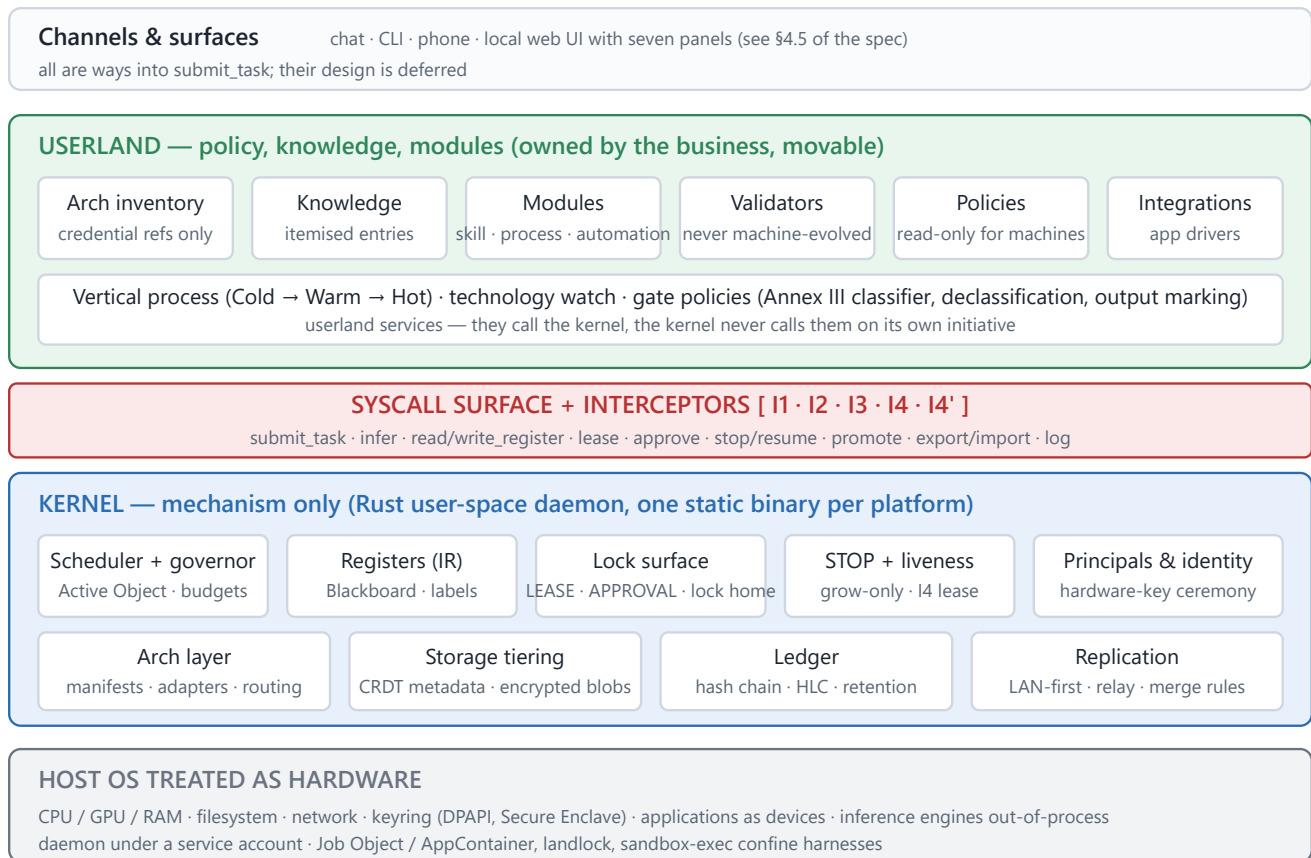


Figure 3 — The boundary. Userland reaches arches, registers and approvals only through the syscall surface; interceptors enforce the invariants on every call. Pattern names: Blackboard, Strategized Locking, Interceptor, Extension Interface, Component Configurator, Active Object, Half-Sync/Half-Async, Kappa.

4. Registers: the task-state IR as a blackboard

A register set holds the state of one task in a model-agnostic form: goal, constraints, evidence with origin taint, decisions, open questions, artefacts by hash, and **classification labels**. Every hand-off between arches goes through the IR — never through one model's native transcript. Verified fact behind this: provider-side reasoning state (e.g. Anthropic thinking blocks) is bound to the producing model and dropped by any other; kernel-owned state is what makes model swapping mid-task possible.

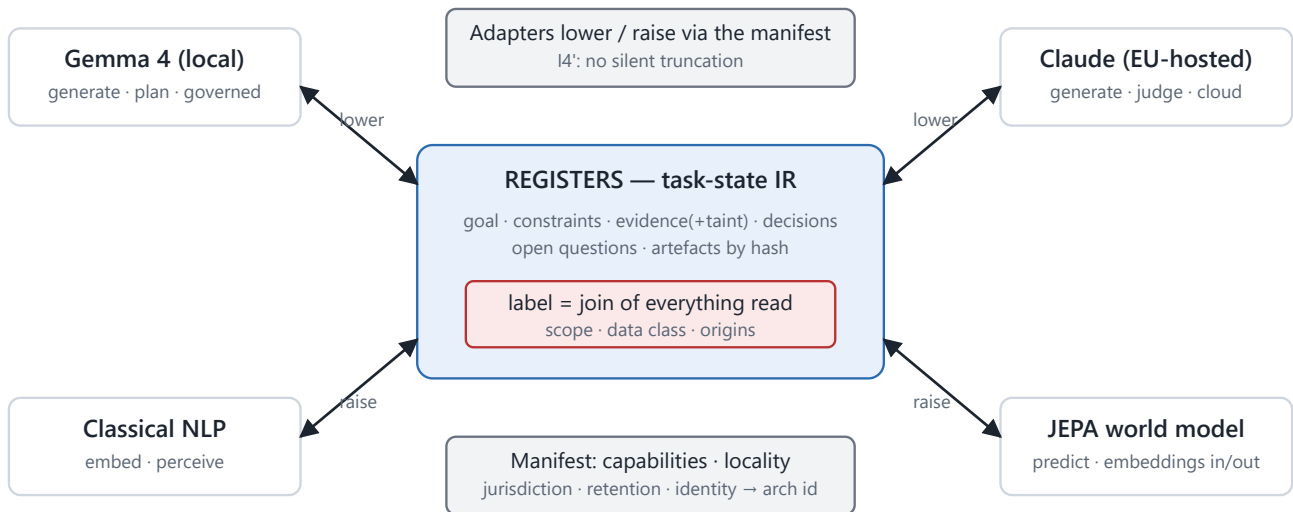


Figure 4 — Blackboard: the arches are knowledge sources; the IR is the only common currency. Topology (hierarchy or not) is policy.

Arch identity is content-addressed. Weights hash, engine + version, backend, quantisation, KV-cache type, threads, batch, sampling, seed — any change is a new arch id (Component Configurator event). A peer's network-reached model is a distinct arch.

Learning modes per artefact class (policy, visible in the UI): *cloud-assisted*, *derived-only* (frontier arches see only declassified outputs), *local-only* (no machine authoring; memory, search, teaching and imported modules remain).

5. Lock surface and partitions

The original arbitration was "one lock primitive with a holder kind {machine, human}". The adversarial review showed it cannot carry its three uses under partition: exclusivity for an irreversible action is a consensus property, and "human locks are local by nature" is exactly what lets two humans each hold "the" lock. The revised arbitration keeps one syscall surface with a **tagged union**.

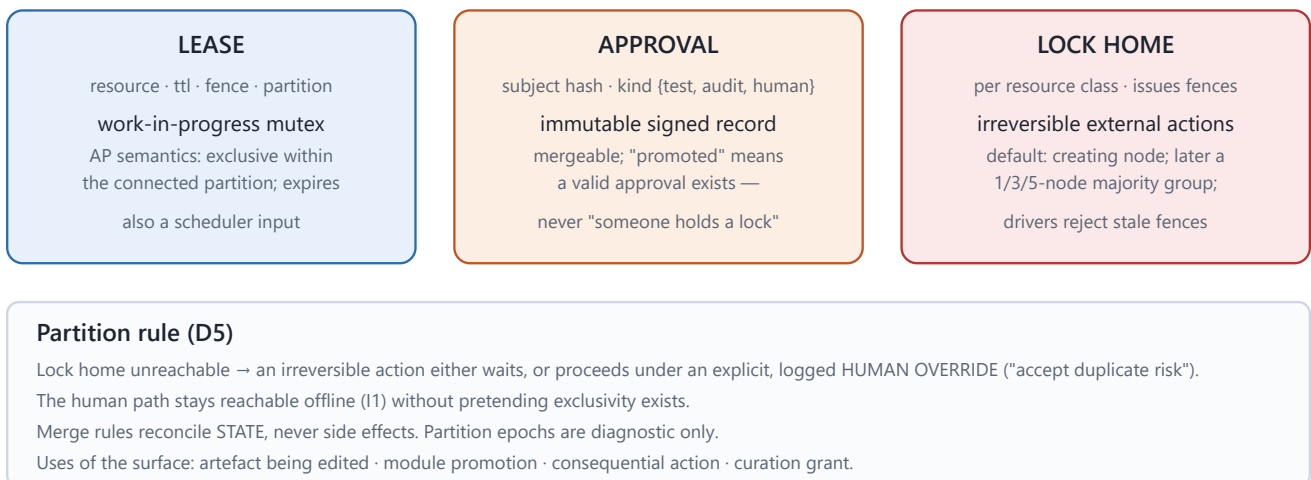


Figure 5 — Three kinds behind one syscall surface (Strategized Locking). Uses: artefact being edited, module promotion, consequential action, curation grant.

6. Invariants and the interceptor chain

Only **decidable** invariants are kernel interceptors. The Annex III classifier and the declassification gate are userland gate policies whose signed verdicts the kernel requires on `promote`, `export` and `import`. All five invariants are property-tested on the SP0 stub kernel.

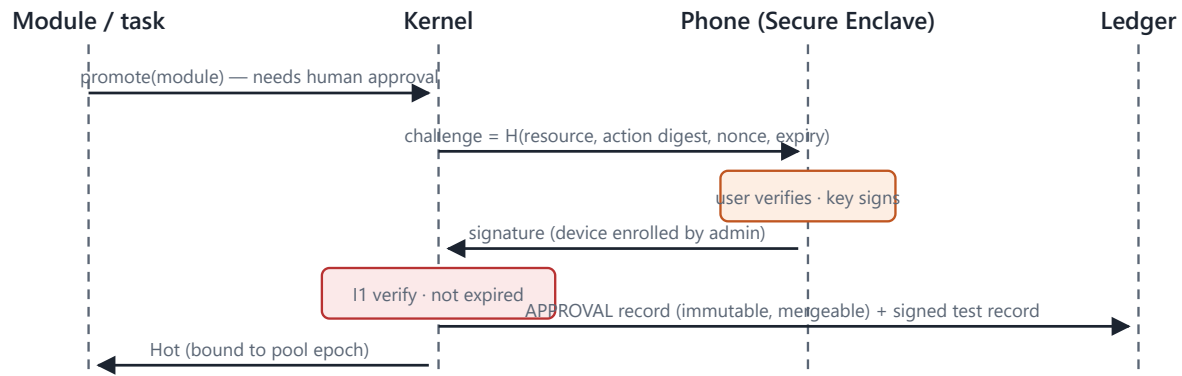


Figure 6 — The interceptor chain on the syscall path.

Inv.	Statement (as restated after review)	SP0 evidence
I1	An approval of kind <i>human</i> can only be produced by a human ceremony; no machine principal can produce or forge it; the human path (STOP, override, approval) is always reachable — no queue, budget or partition can starve it.	property test: no sequence of machine syscalls yields a human approval; human STOP always accepted
I2	No object whose classification exceeds a principal's or arch's effective clearance is ever projected into its reachable set. Every object written by a task inherits the join of the labels it read; declassification is explicit and measurable.	property test: no register label ever reaches an arch above its clearance
I3	A merge never silently discards a write; conflicts are resolved by a declared rule or surfaced, and logged. Logged shred events are the one sanctioned deletion.	property test: every write survives any merge order (kept or surfaced)
I4	Machine principals run non-human-initiated automations only while an autonomy liveness lease, renewed by a human-capable node, is unexpired and no STOP is active in scope.	property test: no automation after lease expiry; STOP blocks until RESUME cites it
I4'	Every adapter reports its real context budget; <code>infer</code> refuses an over-budget lowering or applies a logged structural projection — never a silent truncation.	property test: projected $\Leftrightarrow$ ledger entry exists

7. Human ceremony, STOP and liveness

No syscall accepts a caller-asserted principal; the kernel derives it from the authenticated channel. "Human" is a **hardware-key ceremony** bound to one action digest, short-lived. STOP needs presence only; authorising writes need verification. The phone is an approval channel, never a lock home.



STOP and liveness (I4)

STOP = grow-only set of signed events; RESUME is valid only if it cites the STOP it lifts; "stopped" is a pure function of causal state. Presence suffices — any enrolled device can STOP; authorising writes need user verification on the key.

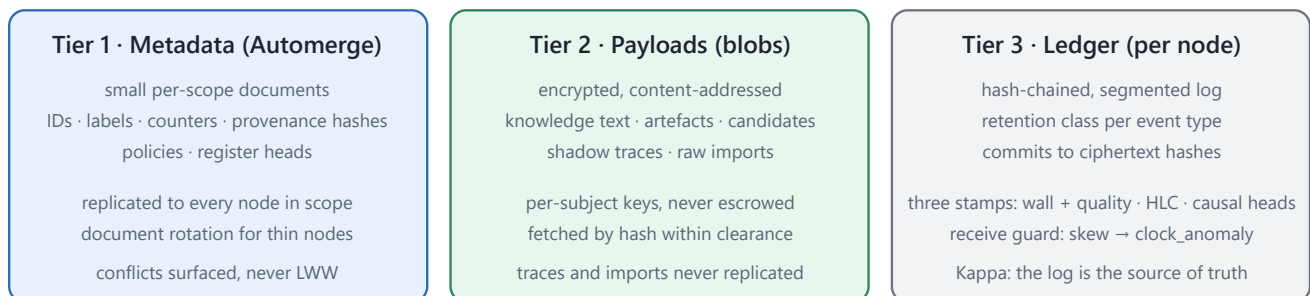
Liveness lease: a node cut off from every human-capable node stops automating when the lease lapses (TTL per autonomy profile). This is what makes AI Act Art. 14(4)(e) "halt in a safe state" true rather than declared.

Devices: enrolment and revocation under the business admin key on a full node (ticket + out-of-band code + admin approval); revocation propagates and wins merges.

Figure 7 — The human ceremony for a promotion, and the STOP/liveness primitives.

8. Storage tiering and the ledger

The original arbitration "Automerge for all mergeable state" was narrowed: full operation history is permanent in a CRDT, tombstones erase nothing, and secrets or traces would replicate to every phone. Three tiers now.



Erasure (GDPR Art. 17)

= a signed SHRED event (merges like STOP) + tombstone: every blob under that subject key becomes unreadable everywhere. The Cold pool is node-local and ephemeral. Secrets live only in the OS keyring of the node that holds the credential; the inventory holds a reference and a fingerprint, and infer() routes to the holding node.

Figure 8 — Storage tiering (D7). SP0 encodes the semantics; SP1 adds real encryption and Automerge.

9. Distribution: partitions and merge

Keep working offline, merge on reconnect, with merge rules per resource class (LAN / VLAN / WAN). Policies merge most-restrictive-wins; STOP, shred and human-override events win merges; "current Hot version" is a set resolved most-restrictive, never a scalar last-writer-wins. Sovereignty restated: **no third party ever holds plaintext or performs inference**; connectivity may use untrusted, self-hostable relays that see ciphertext only.

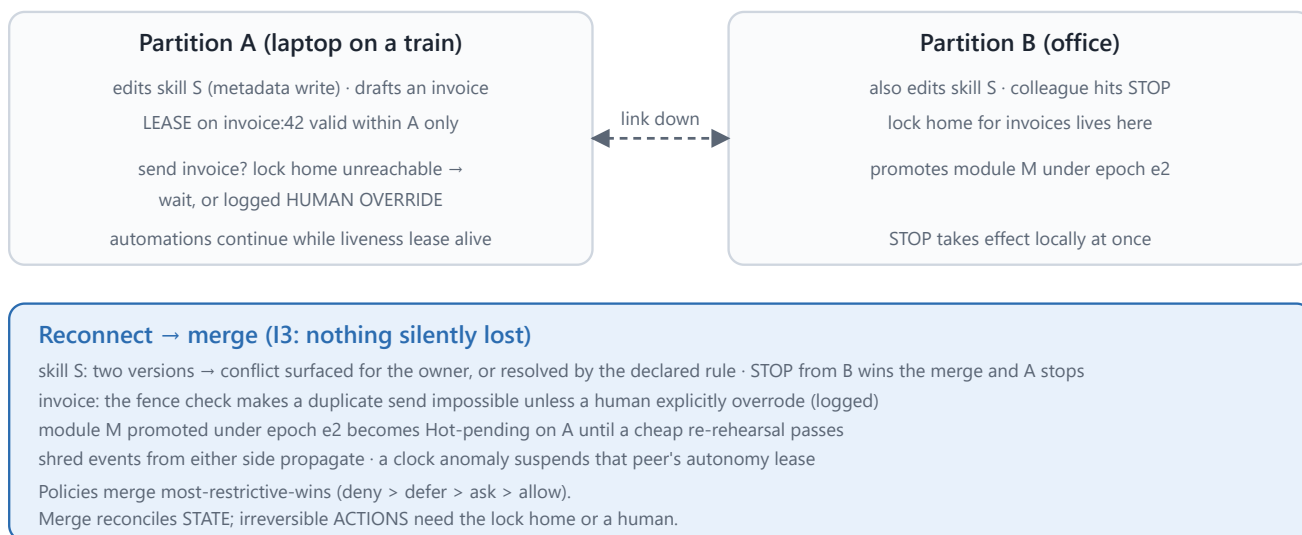


Figure 9 — Partition behaviour (D3, D5, I3, I4).

10. Userland object model

Six families; all versioned, mergeable, carrying a classification label and a provenance manifest (content hash, lineage, signer, arch compatibility, cost/applicability contract, origin taint summary).

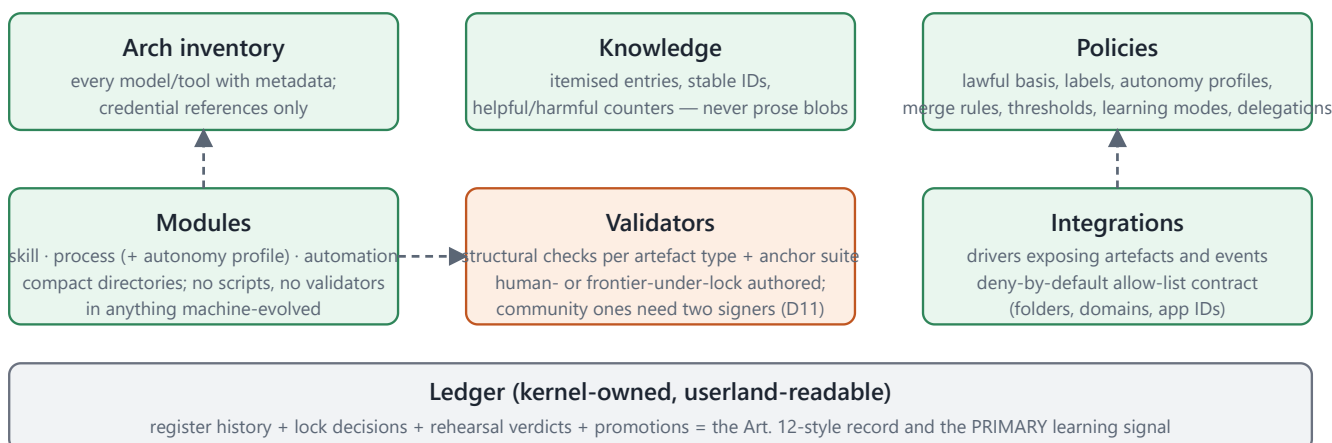


Figure 10 — The six userland families and the ledger they all write to.

11. The vertical process: Cold → Warm → Hot

The literature is unambiguous: self-authored skills are net negative without a hard verifier; success-only telemetry never yields a useful skill; ungated accumulation collapses and rollback recovers little. So the loop is **sparse search with the gate at the front**.

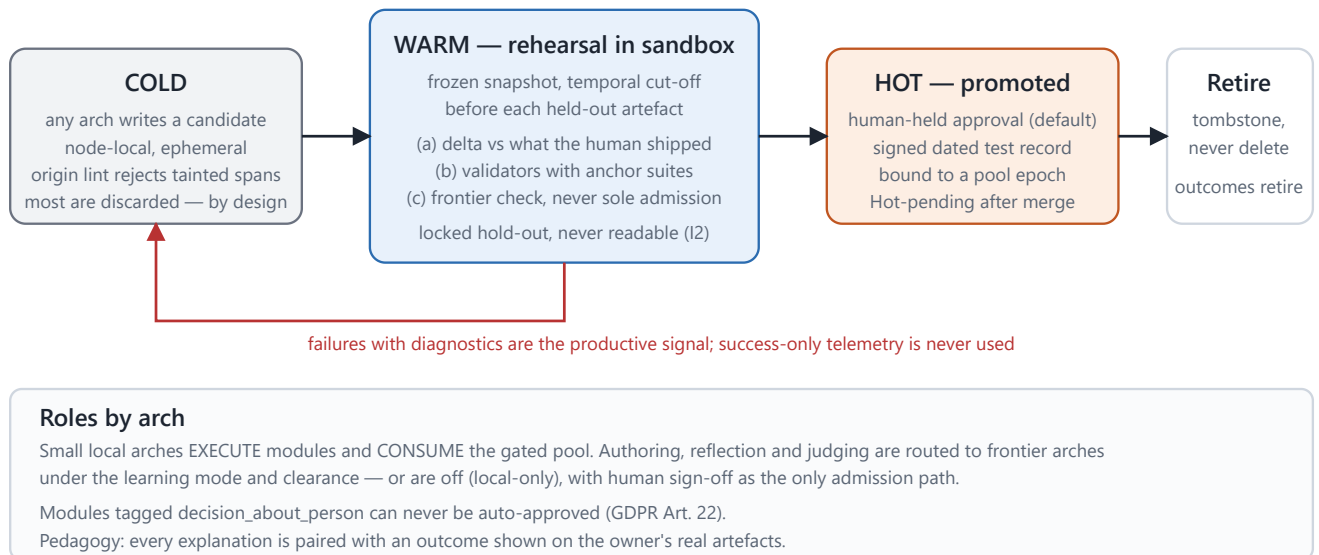


Figure 11 — The gated lifecycle. Promotion is the mutation-gating approval held by the owner: the pedagogic, delegation and oversight surface at once.

12. Learning signals and the threshold

Phase 1 signals (decided): *shadowing* and *technology watch*; explicit teaching later. Shadowing captures, within a user-chosen allow-list, artefact created/modified with diff, tool used, duration — no keystrokes, screenshots or page text outside the allow-list; special-category data dropped at ingest. Every trace carries an origin taint {owner-authored, owner-shipped, third-party-inbound, web} and a data class {own, third-party-mandated, unknown} assigned by source. Traces are personal-scope and non-exportable; only derived, declassified candidates move. Technology-watch candidates carry taint *web* and reach Hot only through human approval.

Threshold (decided, then made two-dimensional): data maturity (volume, artefact-type diversity, lawful-basis coverage) AND verifier readiness per artefact type. VerticalAI sets a floor at which the process fires once automatically for ready artefact types; the owner then owns and adjusts the threshold and is signalled before later runs.

verifier readiness per artefact type	verifier, little data: waits for maturity	first run fires automatically at the floor, then owner-adjusted, signalled before each run
	nothing fires (baseline only)	data, no verifier: excluded from auto run, shown as "needs a validator or N human decisions"
data maturity (volume · diversity · lawful basis)		

Figure 12 — The two-dimensional threshold.

13. Scopes and the curated federation

Three value scopes; each transition is a promotion with the same gates. The vertical scope **is part of the product**: "the first P2P runtime that brings value and adoption to businesses, local-first". The trust model is a **curated federation** that decentralises progressively.

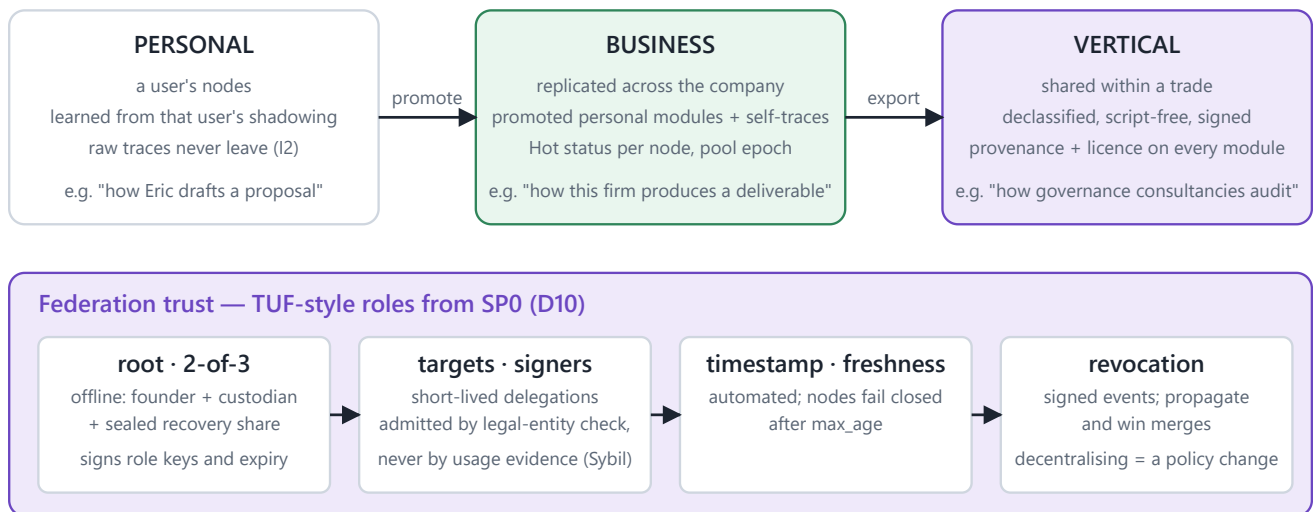


Figure 13 — Scopes (D9) and the curated federation (D10). The end state — businesses vetting each other with no central party — is a goal revisited when Sybil-resistant admission exists.

14. Compliance mapping (corrected after review)

Obligation	Applies	Mechanism in VerticalAI
GDPR lawful basis for shadowing	now	Sole trader: consent. Workplace: Art. 6(1)(f) legitimate interest + Art. 88 national rules + mandatory DPIA (Art. 35(3)(a)); works-council/CSE attestation object (FR: L2312-38). Employee consent is not a lawful basis (EDPB 05/2020).
e-Privacy Art. 5(3)	now	Browser/app drivers need the device user's consent; sole-trader self-install fits the requested-service exception.
GDPR Art. 28	now	Client artefacts are processed as processor; data class <i>third-party-mandated</i> restricts learning to local/on-prem arches absent documented instructions.
GDPR Art. 17 erasure	now	Per-subject keys + signed shred events (§8).
GDPR Art. 22	now	<code>decision_about_person</code> modules cannot be auto-approved; ledger entry is the intervention record.
AI Act Art. 50 transparency	2 Aug 2026	Output-marking gate for every <i>generate</i> arch: C2PA/IPTC for media, provenance record for text; interaction disclosure in channels.
AI Act Art. 25(2)	now	Instructions for use + machine-unwritable policies declare the system "is not to be changed into a high-risk AI system"; Annex III classifier runs at <code>propose_candidate</code> , <code>promote</code> , <code>import</code> and on the shadowing configuration itself (Annex III 4(b)); Art. 6(3) derogation documented where used.
AI Act Arts. 9, 12, 14, 26, 43(4) / Art. 3(23)	conditional; Annex III from 2 Dec 2027 (Reg. 2026/1744)	Ledger, STOP + liveness lease + hardware-key approvals, pre-determined-change envelope per trade. The design prevents high-risk use by default; the mechanisms exist so a deployer who chooses a high-risk purpose can comply.

Cyber Resilience Act	reporting from 11 Sep 2026; main application 11 Dec 2027	Approval-gated, logged self-modification; SBOM incl. weights and runtimes; default-on security updates with opt-out; 24h/72h/14d reporting via ENISA (opt-in security-event telemetry); 5-year support; identity/key/signer component packaged separately (class I "identity management"); "operating system" dropped from external naming.
Product Liability Directive 2024/2853	9 Dec 2026	Software is a product; signer and vendor liability documented in federation terms.
NIS 2	customer-side	Logs, access control and incident evidence supplied to essential/important entities.

All citations are to be re-verified against the consolidated Official Journal text before any external publication.

15. Decision log

Design decisions (spec §2)

#	Decision	Notes
D1	Kernel owns only orchestration, registers, locks; everything else is userland	mechanism in kernel, policy in userland
D2	Model = ISA; heterogeneous arches behind manifests; kernel is model-agnostic	any AI system: local LLMs, cloud LLMs, world models, classical NLP
D3	P2P multi-node, one shared userland per business, no cloud dependency	offline-first, merge on reconnect
D4	Locks: tagged union (LEASE / APPROVAL / lock home) behind one syscall surface	revises "one lock primitive" after review
D5	Irreversible actions: per-resource-class lock home + logged human override when unreachable; quorum groups later	founder's choice among the review's options
D6	Human principal = hardware-key ceremony; phone is an approval channel, never a lock home, persists nothing	
D7	Storage tiering: Automerge for small metadata only; encrypted content-addressed blobs; per-node hash-chained ledger	narrows "Automerge for mergeable state"
D8	Learning signals phase 1: shadowing + technology watch; explicit teaching later	
D9	Three value scopes: personal → business → vertical; the vertical scope is part of the product	
D10	Curated federation: founder as explicit root (2-of-3), signer admission by legal-entity check, progressive delegation	"no central party" is the end-state goal
D11	Validators are a separate object class excluded from machine evolution	
D12	Sole-trader mode first; workplace mode deferred until DPIA/LIA/works-council objects exist	
D13	EU-hosted frontier arch as first-class adapter ("your data, your jurisdiction")	Anthropic first-party API has no EU inference geography

D14	Spec-first, then a Rust user-space kernel; SP1 demo gated on isolation and code signing	Go runner-up; Python rejected for the kernel
D15	Commercial first; kernel open-sourced later	

Founder arbitrations during brainstorming (22 Sep 2026)

Question raised	Arbitration
What does the kernel own?	Micro-architecture analogy: orchestration, registers (memory), lock/semaphore concept applied to AI. The rest is userland — that is what makes the product movable.
Which lock semantics?	All three: resource mutual exclusion + resource allocation for the scheduler; mutation gating; human-held semaphores. Human-in-the-loop is a concept CARS puts on top: policy in userland, but the kernel keeps a primitive guaranteeing it is available at all times.
Which models?	None specific; one or a combination; open-source local and big providers (Fable cited) are both legitimate arches. Any AI system — JEPA world models, System-1-style, NLP, future paradigms — must be characterisable.
Decentralised how?	Several kernels on several nodes sharing one userland, no cloud, P2P propagation with management on top — the same value for a one-person company and a large enterprise.
Partition behaviour?	Keep working, merge on reconnect with configurable merge rules (LAN/VLAN/WAN) — the most evolutive option.
Where does Claude Code sit?	Corrected: the OS schedules Claude Code when needed; harnesses are opaque userland processes under a lease.
What is a node?	Computation as close to the system as possible with a minimum performance floor; a configurable share of machine resources; native on Windows (most businesses), Linux and macOS; must live on the same host as the applications it integrates.
Which learning signals first?	Shadowing and technology watch; explicit teaching afterwards. The owner must understand what is built and keep decision-making and personalisation.
Is the vertical scope part of the product?	Yes — a wider end point in clients and an innovation: the first P2P runtime bringing value and adoption to businesses, local-first.
Trust model of the vertical scope?	VerticalAI curates first; curation is delegated to businesses with enough usage; end state: businesses checking each other and allowing connection ("vertical partnership"). Curated federation with the founder as signer for now.
First trade?	Freelancers and SMEs offering services/products — not the founder's audit practice — because SME owners lack the skills and tools to find relevant use cases and many already sell AI-made work.
Week-one behaviour?	CARS-style register first: inventory of models and tools with all metadata (conversations, usage, licences, master prompts) into one local workspace; a threshold with a floor that fires the first run automatically, then owner-adjusted; all trades, not just quotes and emails.

Founder decisions after the adversarial review (23 Sep 2026)

#	Decision
R1	Locks under partition: lock home + logged human override for irreversible actions; quorum groups later.

R2	Licence: commercial first to build activity; the kernel is open-sourced later to democratise the commercial offering.
R3	Workplace mode: sole traders first; the 12-person agency waits for DPIA/LIA/CSE objects, employee-held trace keys and k-anonymous deltas.
R4	Human identity: hardware-key ceremony (Windows Hello/TPM, Secure Enclave/StrongBox); phone = approval channel; enrolment friction accepted.
R5	Local-only businesses: an EU-hosted frontier arch is a first-class adapter so "jurisdiction: EU" is selectable; the product states plainly that fully-local deployments get no machine authoring.
R6	Root of trust: the founder under a 2-of-3 threshold key; signer admission anchored to a legal-entity check, never usage evidence.
R7	Validators leave the machine-evolution loop entirely.
R8	Naming and sequencing: "AI operating system" dropped externally; the SP1 Windows demo is gated on service-account isolation, harness confinement and code signing rather than "demo first, harden later".

16. Hypotheses register

Everything below is a bet the roadmap must confirm or refute. "Evidence" is what we already know; "Test" is the sub-project and observable that settles it.

#	Hypothesis	Evidence today	Test	Status
H1	A model-agnostic task-state IR lets heterogeneous models cooperate on one task without losing state.	Provider reasoning state is model-bound (verified); AIOS/AgentOS keep kernel-side context managers.	SP1 demo: Gemma 4 + Claude Code on one task with a mid-task arch swap; IR round-trip loses no decision.	to validate
H2	Small local models (4–9B) can execute compact gated modules and consume a gated pool usefully.	Curated compact skills + 16.6 pp; gated pool consumed by a small MoE + 16 pp; 7B models weak on long multi-app chains.	SP3: module execution success on the owner's artefact types with Gemma 4 vs a frontier arch.	literature
H3	Machine authoring of modules needs a frontier arch; local-only businesses get no machine authoring.	No source shows a 4–9B model authoring or judging skills; self-authoring net negative without a verifier.	SP3: candidate acceptance rate by authoring arch class.	literature
H4	The delta-to-shipped verifier, with temporal cut-off and a locked hold-out, is a valid admission signal.	Review: leaky and Goodhart-able as first specified; fixed by cut-off + hold-out; still confounded by edits made for non-quality reasons.	SP3: false-discovery rate on hold-out; correlation of delta with owner accept/reject decisions.	to validate
H5	SME owners will review a proposals inbox when it shows outcomes on their own artefacts.	"Bot-PR neglect" raised and refuted; explanations without outcome feedback worsen judgment ($g = -0.31$).	SP3 pilot: proposals reviewed within 7 days; promotion rate; retention after 30 days.	to validate
H6	The baseline inventory is valuable on its own in week one.	Review confirmed value only if real content is imported (browser capture driver, receipts importer moved to SP2).	SP2 pilot: owner uses the workspace daily for a week without any self-improvement.	to validate

H7	Merge-on-reconnect with surfaced conflicts is acceptable to SMEs.	Local-first literature; I3 property-tested; conflicts must be rare enough not to annoy.	SP4: conflicts per week per business in the pilot; time-to-resolve.	to validate
H8	An EU-hosted frontier arch satisfies sovereignty-minded customers.	Anthropic first-party API: US/global only, 30-day retention; Bedrock/Vertex EU exist.	SP2-3: share of pilots choosing EU-hosted vs local-only vs US cloud.	to validate
H9	A curated federation can bootstrap a vertical before decentralising.	Pure-P2P skill patches leak ~5% entities; usage-evidence reputation is Sybil-attackable; TUF roles are standard.	SP5: first trade with ≥ 3 businesses importing signed modules; zero revocations for provenance failure.	to validate
H10	The hardware-key ceremony's enrolment friction is acceptable.	Windows Hello/TPM and Secure Enclave APIs exist; no user data yet.	SP4: enrolment completion rate; approvals per day per owner.	to validate
H11	A two-dimensional threshold (maturity $\times$ readiness) fires usefully rather than never.	Review: gating on volume alone manufactures false Hot modules; readiness needs a validator or N human decisions.	SP3: time-to-first-run per artefact type; fraction of types that ever become ready.	to validate
H12	Show-outcomes-not-explanations improves owner judgment.	74-study re-analysis: explanations + outcome feedback $g = +0.30$; LLM explanations gave zero human gain.	SP3: A/B of proposal cards with/without explanation text.	literature
H13	Rust + spec-first does not stall the project.	SP0 delivered in one day on contracts; toolchain friction on Windows (MSVC required) documented.	SP1 delivered within its planned window with the demo gate met.	to validate
H14	A user-space daemon can enforce I2 on Windows (service account, keyring, harness confinement).	Review: unenforceable without these; Job Object / AppContainer exist; Claude Code MCP mode disclaims enforcement.	SP1 gate: red-team attempt by a confined harness to read the store or reach an arch directly fails.	known risk
H15	Shadowing can be GDPR-defensible for a sole trader and later for workplaces.	Consent valid for sole traders; workplaces need Art. 6(1)(f) + Art. 88 + DPIA; shadowing itself may be Annex III 4(b).	SP3: DPIA template accepted by counsel; Annex III classifier verdict on the shadowing configuration.	known risk
H16	The three-tier storage keeps replicas bounded and erasure real.	Review: CRDT-only storage cannot; tiering is standard practice in local-first systems.	SP4: replica size over 90 days; erasure verified unreadable on every node within one sync.	to validate
H17	"Orchestration runtime" naming avoids CRA class I while the identity component is assessed separately.	A user-space daemon is not an OS; identity/key/signer matches class I "identity management".	SP5: CRA classification memo reviewed by counsel before market placement.	known risk
H18	Technology watch produces candidates worth gating.	Review: an inbound injection channel with no ground truth; only human approval can admit.	SP3: share of tech-watch proposals promoted by owners.	to validate

17. Roadmap and SP0 status

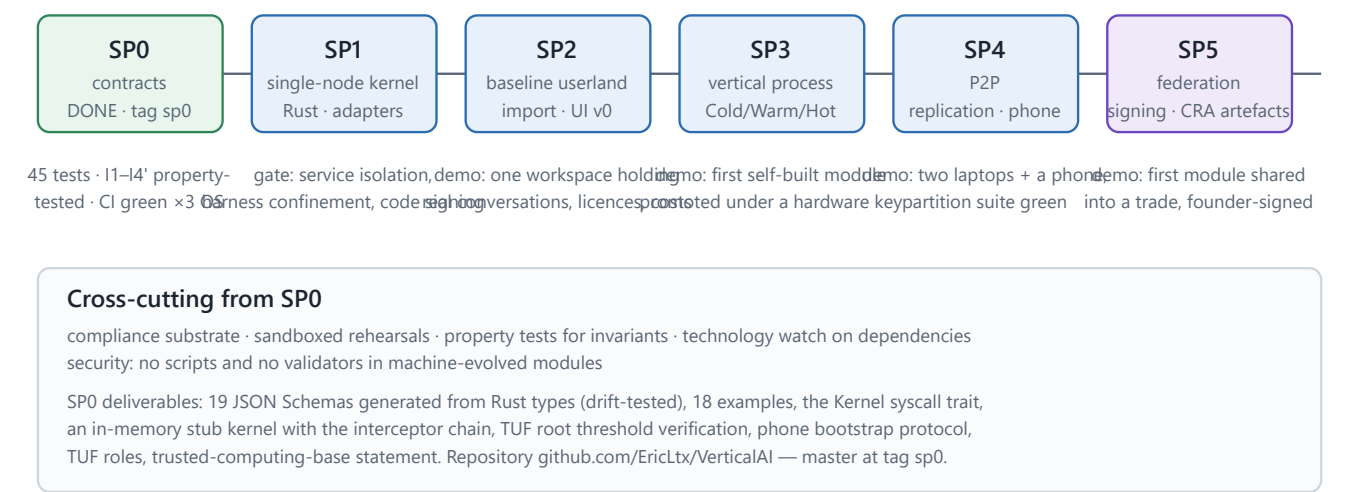


Figure 14 — Sub-projects, gates and demos (spec §6).

18. Open questions and deferred items

Item	State
Channels (chat / CLI / phone UX)	Deferred; they are ways into <code>submit_task</code> .
Explicit teaching as a learning signal	Phase 2, after shadowing and technology watch.
Workplace-mode objects (DPIA, LIA, CSE attestation, employee-held trace keys, k-anonymous deltas)	Deferred with R3; move into SP0/SP2 if a multi-person business enters scope before SP3.
Quorum lock homes (1/3/5-node majority)	After SP4; default is the creating node + human override.
"No central party" admission mechanism for the federation	Goal; revisited when Sybil-resistant admission exists.
JEPA-style and classical-NLP adapters	Manifests allow them; no adapter planned before SP3.
On-device fine-tuning from one business's data	Not planned: no evidence it helps at 4–9B; the small-model literature itself asks for 10k–100k examples.
Statistical design of the delta verifier (multiplicity at the Cold gate)	To be specified in SP3's plan with a locked hold-out and false-discovery control.
Which business processes have a verifier at all?	Those without one may stay permanently un-evolvable or use human sign-off as their only signal — to be measured per artefact type in SP3.
Pedagogic loop when outcomes arrive weeks later, elsewhere, or never	Outcomes retire modules; they never admit them. Rehearsal-only outcomes are shown as such.

Generated from the design spec, the literature brief and the adversarial review on 23 September 2026. This document records arbitrations; the specification remains the normative text.